

VPC Connection Through Transit Gateway.

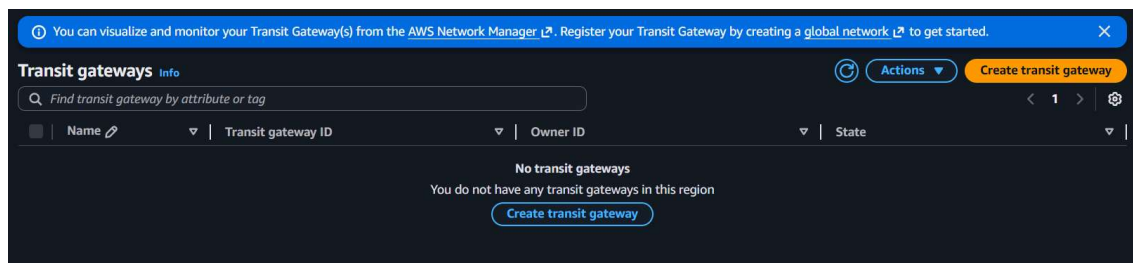
1. What is AWS Transit Gateway?

AWS Transit Gateway is a **network transit hub** that connects multiple **Amazon Virtual Private Cloud (VPCs)** and **on-premise networks** using a **centralized router model**.

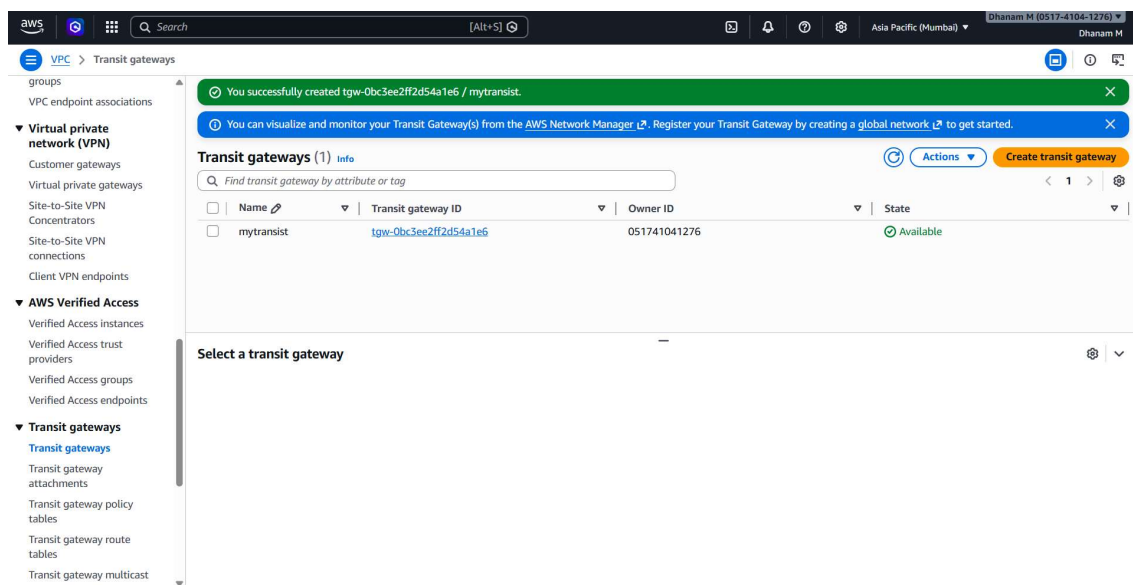
Instead of connecting VPCs individually using many **VPC Peering** connections, Transit Gateway acts like a **hub** where all networks connect to one central point.

Think of it like:

- **Transit Gateway = Router**
- **VPCs / VPN / Direct Connect = Networks connected to that router**



I have to create Transit Gateway for my three VPC's.



And then I have to create an three Transit Gateway Attachments.

TRANSIST – 1

aws    Search [Alt+S]

VPC > Transit gateway attachments > Create transit gateway attachment

Create transit gateway attachment [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

Transit gateway ID [Info](#)

Attachment type [Info](#)

VPC attachment

Select and configure your VPC attachment.

☒ DNS support [Info](#)

☒ Security Group Referencing support [Info](#)

☐ IPv6 support [Info](#)

☐ Appliance Mode support [Info](#)

VPC ID
Select the VPC to attach to the transit gateway.

Subnet IDs [Info](#)
Select the subnets in which to create the transit gateway VPC attachment.

TRANSIST – 2

aws    Search [Alt+S]

VPC > Transit gateway attachments > Create transit gateway attachment

Create transit gateway attachment [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
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Transit gateway ID [Info](#)

Attachment type [Info](#)

VPC attachment

Select and configure your VPC attachment.

☒ DNS support [Info](#)

☒ Security Group Referencing support [Info](#)

☐ IPv6 support [Info](#)

☐ Appliance Mode support [Info](#)

VPC ID
Select the VPC to attach to the transit gateway.

Subnet IDs [Info](#)
Select the subnets in which to create the transit gateway VPC attachment.

TRANSIST –3

Create transit gateway attachment [info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
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Transit gateway ID [info](#)

Attachment type [info](#)

VPC attachment
Select and configure your VPC attachment.

☒ DNS support [info](#)

☒ Security Group Referencing support [info](#)

☐ IPv6 support [info](#)

☐ Appliance Mode support [info](#)

VPC ID
Select the VPC to attach to the transit gateway.

Subnet IDs [info](#)
Select the subnets in which to create the transit gateway VPC attachment.

Transit gateway attachments (3) [info](#)

☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
☐	transist3	tgw-attach-039f4046b7baaddf	tgw-0bc3ee2ff2d54a1e6	Available	VPC	vpc-0a7805f4f020e4b21
☐	transist	tgw-attach-08d6d5bbddcfd305	tgw-0bc3ee2ff2d54a1e6	Available	VPC	vpc-0b9db0ae6f01c7fa17
☐	transist2	tgw-attach-09d7b6c2c19b19f85	tgw-0bc3ee2ff2d54a1e6	Available	VPC	vpc-07fc81f77dc4bd48c

Select a transit gateway attachment

I have created the VPC1,VPC2,VPC3 for three different VPC's.

Your VPCs (4) [info](#)

Last updated 8 minutes ago

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 C
☐	-	vpc-0a3751e43faed7be1	Available	-	-	Off	172.3
☐	vpc-2	vpc-0721c4160907c095c	Available	-	-	Off	12.0.C
☐	vpc-3	vpc-01c1394144326851f	Available	-	-	Off	13.0.C
☐	vpc-1	vpc-038015eda77c2e886	Available	-	-	Off	11.0.C

I have created the three TGA for three different VPC's.

Transit gateway attachments (3) [info](#)

☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
☐	attachment-3	tgw-attach-01d879a49851eff79	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-01c1394144326851f
☐	attachment-2	tgw-attach-069390d9696a30403	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-0721c4160907c095c
☐	attachment-1	tgw-attach-0d7fd316f97049c23	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-038015eda77c2e886

Then I have edited routes for all the VPC's.

Route-1

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Q local	-	No	CreateRoute
Q 11.0.0.0/16	Internet Gateway	-	No	CreateRoute
Q 12.0.0.0/16	Q igw-0cff4788486eef537	-	No	CreateRoute
	Transit Gateway	-	No	CreateRoute
	Q tgw-0bc3ee2ff2d54a1e6	-	No	CreateRoute
	Transit Gateway	-	No	CreateRoute
	Q tgw-0bc3ee2ff2d54a1e6	-	No	CreateRoute

Add route

Cancel

Preview

Save changes

Route – 2

Edit routes

Destination	Target	Status	Propagated	Route Origin
11.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Q local	-	No	CreateRoute
Q 10.0.0.0/16	Internet Gateway	-	No	CreateRoute
Q 12.0.0.0/16	Q igw-04943f9029237c7c	-	No	CreateRoute
	Transit Gateway	-	No	CreateRoute
	Q tgw-0bc3ee2ff2d54a1e6	-	No	CreateRoute
	Transit Gateway	-	No	CreateRoute
	Q tgw-0bc3ee2ff2d54a1e6	-	No	CreateRoute

Add route

Cancel

Preview

Save changes

Route – 3

Edit routes

Destination	Target	Status	Propagated	Route Origin
12.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Q local	-	No	CreateRoute
Q 10.0.0.0/16	Internet Gateway	-	No	CreateRoute
Q 11.0.0.0/16	Q igw-047fe506864e3a68f	-	No	CreateRoute
	Transit Gateway	-	No	CreateRoute
	Q tgw-0bc3ee2ff2d54a1e6	-	No	CreateRoute
	Transit Gateway	-	No	CreateRoute
	Q tgw-0bc3ee2ff2d54a1e6	-	No	CreateRoute

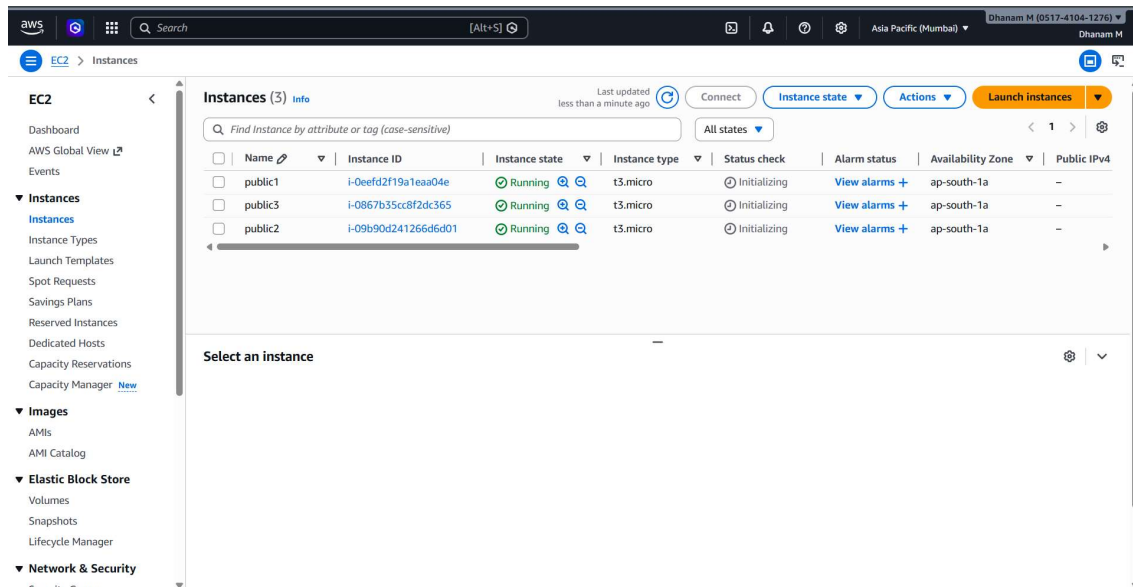
Add route

Cancel

Preview

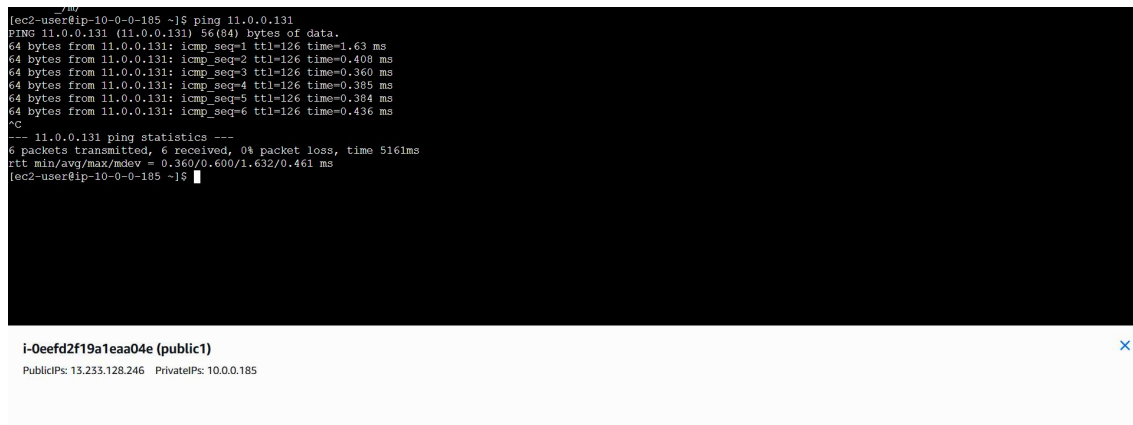
Save changes

The created 3 INSTANCES

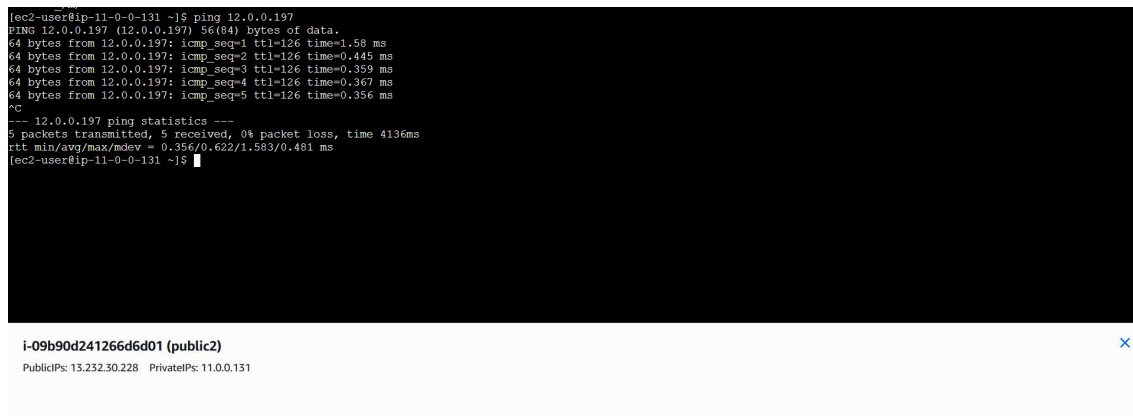


Then I have checked in all the instances having internet using the ping command.

Public1



Public2



Public3

```
lec2-user@ip-12-0-0-197 ~]$ ping 10.0.0.185
PING 10.0.0.185 (10.0.0.185) 56(84) bytes of data.
64 bytes from 10.0.0.185: icmp_seq=1 ttl=126 time=1.39 ms
64 bytes from 10.0.0.185: icmp_seq=2 ttl=126 time=0.363 ms
64 bytes from 10.0.0.185: icmp_seq=3 ttl=126 time=0.361 ms
64 bytes from 10.0.0.185: icmp_seq=4 ttl=126 time=0.347 ms
^C
--- 10.0.0.185 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3058ms
rtt min/avg/max/mdev = 0.347/0.614/1.385/0.445 ms
lec2-user@ip-12-0-0-197 ~]$
```

i-0867b35cc8f2dc365 (public3)

PublicIps: 13.126.140.154 PrivateIps: 12.0.0.197